Warehouse Anomaly Detection & Tail-Risk Stress Testing

Illustrative Framework & Technical Report — Undergraduate Research Project
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GitHub: https://github.com/Shridathj/Warehouse-Anomaly-Detection-Tail-Risk-Stress-Testing | **Kaggle:** https://www.kaggle.com/code/prnavjoshi/warehouse-anomaly-detection-stress-testing

■ CRITICAL SCOPE NOTE — READ FIRST

This is an undergraduate research project created under real constraints: no access to proprietary WMS telemetry, no real 2025 fulfillment logs, and only the 2010–2011 UCI Online Retail dataset publicly available. All delay distributions are fully synthetic. All parameters (dragon frequency, delay means, holding rates, etc.) are educated guesses derived from publicly available summaries of the 2025 WERC DC Measures and CSCMP State of Logistics reports — not the full proprietary datasets. Financial estimates are Monte Carlo outputs under stated assumptions and are directional/illustrative only. This framework is NOT validated on real warehouse data and must NOT be used for operational or financial decisions without independent validation on current, real-world data. Statistical power of backtesting is inherently low due to limited data length.

1. Abstract

This undergraduate project develops an **illustrative** framework for quantifying tail-risk exposure from extreme high-value "dragon" orders (99th-percentile transactions) when warehouse fulfilment service levels degrade from the 99th to the 95th percentile. Using the only credible public transactional dataset available (UCI Online Retail, 2010–2011: 541k rows, 3,665 SKUs, £8.9M gross revenue over 374 days), two complementary scenarios were modelled: **gross** (maximum exposure) and **netted** (realistic after cancellations). Realistic delays were simulated using **fully synthetic** log-normal distributions with parameters derived from public summaries of the 2025 WERC DC Measures Report and CSCMP State of Logistics Report. The end-to-end pipeline integrates extreme-value theory (EVT/GPD), Monte-Carlo simulation (10,000 paths), causal inference (PSM + quantile regression), Hawkes self-exciting processes, and Bayesian Structural Time Series forecasting. Purged expanding-window backtesting was performed with Kupiec and Christoffersen tests. All results are **assumption-driven and directional**. Maintaining 99th-percentile SLA on high-value orders reduces modelled exposure to near zero under the stated assumptions.

2. Executive Summary

The Warehouse Anomaly Detection & Tail-Risk Stress Tester provides a **reproducible technical pipeline** for stress-testing warehouse tail risk under data and access constraints typical of undergraduate research. Under **base-case assumptions**, relaxing service levels from the 99th to 95th percentile on high-value dragon orders generates modelled annual exposure of approximately **£77,000 (netted scenario)** to **£268,000 (gross scenario)**. Sensitivity analysis on key parameters (dragon frequency, delay means, holding rate, bias exponent) produces ranges of roughly **£55k–£320k (netted)** and **£180k–£420k (gross)**. 99th-percentile fulfilment on dragon orders keeps modelled exposure near zero in simulation. Backtesting on the limited 374-day public sample shows 2/33 violations in the netted scenario (Kupiec p=0.79, Christoffersen p=0.61) — acceptable but with **low statistical power**. The gross scenario shows 0 violations, as expected under heavier-tailed assumptions and higher thresholds. **This work demonstrates end-to-end implementation of advanced quantitative methods** (EVT, Hawkes, BSTS, Monte Carlo, purged backtesting) suitable for further development with real WMS data. It does **not** constitute validated operational guidance.

3. Data, Scope & Assumptions

Primary Data Source: UCI Online Retail dataset (Chen, 2015) — the only large-scale, credible public transactional retail dataset available. 541,000 raw rows, 3,665 SKUs, £8.9M gross revenue over 374 days (Dec 2010 – Dec 2011). After cleaning (positive quantity & unit price, non-missing CustomerID, removal of miscellaneous codes), two parallel scenarios were created: **Scenario 1 (Gross/Max Exposure)** — no netting of refunds/cancellations; **Scenario 2 (Netted/Realistic)** — full CustomerID–SKU netting to reflect actual fulfilled demand.

Why 2010–2011 data? No more recent large-scale public transactional retail dataset with CustomerID, SKU, quantity, unit price, and timestamp exists in the public domain. Modern WMS data is proprietary. This dataset remains the standard benchmark for retail analytics research despite its age.

Delay Simulation: All delays (normal, surge, dragon) are **fully synthetic** — generated from log-normal distributions with parameters (means, sigmas, clip ranges, dragon bias exponents) derived from **publicly available summaries** of the 2025 WERC DC Measures Report and CSCMP State of Logistics Report, cross-referenced with secondary sources (Yale Materials Handling infographic, public industry benchmarks). No real warehouse telemetry or SLA breach logs were available. See LIMITATIONS for full disclosure of every assumption.

4. Parameter Calibration (Educated Assumptions)

All values below are **conservative assumptions informed by public summaries** of 2025 industry reports. They are not direct extractions from the full proprietary datasets. See Limitations section for detailed sourcing.

Parameter	Scenario 1 (Gross / Max Exposure)	Scenario 2 (Netted / Realistic)	Source / Basis
Annual holding rate	25 % APR	25 % APR	Industry standard (WERC summaries; 20–30 % APR widely cited)
Surge events	18 %	18 %	WERC / CSCMP 2025 capacity & volatility summaries
Dragon-tier frequency	0.032 % (32 bps)	0.025 % (25 bps)	Tail-risk calibration to 99th-percentile orders (assumed)
Dragon bias exponent	1.35	1.20	Heavy-tail concentration adjustment (assumed)
Normal / Surge / Dragon delay mean	25 / 90 / 420 min	25 / 90 / 360 min	WERC on-time delivery & surge metrics (public summaries)
Normal / Surge / Dragon σ	0.50 / 0.65 / 0.85	0.50 / 0.65 / 0.82	WERC cycle-time variability summaries
Normal / Surge / Dragon clip range	5–120 / 30–480 / 180–1,440 min	5–120 / 30–360 / 200–1,440 min	WERC realistic SLA windows (public)
SLA breach threshold (Unfulfilled Dragon)	≥240 min (4 hours)	≥360 min (6 hours)	WERC service-level definitions (public summaries)
SLA bins (On-Time / Minor / Breach / Failure)	[0, 30, 120, 480, ∞]	[0, 30, 120, 360, ∞]	WERC / CSCMP SLA tiering (public)

Table 1: All parameters are conservative assumptions informed by publicly available summaries of 2025 WERC DC Measures Report and CSCMP State of Logistics Report. Full proprietary datasets were unavailable. See Limitations for exact sourcing.

5. Detailed Methodology

The pipeline consists of nine rigorously linked stages. All stages using real data are reproducible from the public UCI dataset. Stages involving delay simulation are fully synthetic and documented in config.py.

1. **Data Ingestion & Pre-processing** — Filtering (positive quantity & unit price, non-missing CustomerID, removal of miscellaneous codes) and CustomerID–StockCode netting for Scenario 2.
2. **Global Diagnostics & Pareto Filtering** — 80/20 rule applied to identify high-value SKUs and customers.
3. **Extreme-Value Tail Modelling (EVT/GPD)** — Hill, moment, GEV estimators and peaks-over-threshold fitting on order value distribution to identify dragon threshold.
4. **Realistic Delay & Anomaly Simulation** — Tiered (normal/surge/dragon) log-normal delays with value-biased dragon assignment. All parameters in config.py; fully synthetic.
5. **Monte-Carlo Value-at-Risk & Expected-Shortfall** — 10,000 annual paths, VaR95/99, ES95. Daily dragon revenue assumed i.i.d. lognormal (clustering handled separately via Hawkes).
6. **Causal Validation** — Propensity-score matching (1:1 caliper 0.005, 50k subsample) + quantile regression (100k subsample, 500 bootstrap replications) on order value vs. delay.
7. **Temporal Dependence & Forecasting** — Hawkes self-exciting process for anomaly clustering + Bayesian Structural Time Series (PyMC/PyStan) with conservative fixed priors.
8. **Purged Expanding-Window Backtesting** — Minimum 180-day training, 30-day step, 14-day purge gap. Kupiec and Christoffersen tests on VaR breaches.
9. **Reporting & Stress Interpretation** — Actionable mitigation dashboard (fast lanes, dedicated safety stock for dragon-tier SKUs).

6. Backtesting Results & Financial Impact (Base Case)

Scenario 1 (Gross — Maximum Exposure): 0 violations out of 34 (0.0%). Kupiec p=1.00, Christoffersen p=1.00. *Note: Heavier loss tail + higher VaR/ES thresholds produce zero breaches by construction. This is expected, not a validation win.*

Scenario 2 (Netted — Realistic): 2 violations out of 33 (6.1%). Kupiec p=0.79, Christoffersen p=0.61. *This is the primary operational model. Violation rate is acceptable; statistical power is low due to only ~1 year of data.*

Base-case point estimates (Monte Carlo ES95 at 95th-percentile SLA):
• Realistic (netted) operations: **£77,459** modelled annual preventable dragon loss.
• Gross view: **£268,153** modelled annual exposure.
• 5-year cumulative (constant conditions): **£387,000 – £1.34 million** (assumption-dependent).

Sensitivity note: Varying dragon frequency (±30%), dragon delay mean (±25%), and holding rate (±5pp) produces ranges of approximately **£55k–£320k (netted)** and **£180k–£420k (gross)**. Full sensitivity tables in repo/results/.

Directional Implications (Not Recommendations): Prioritising 99th-percentile SLA on high-value orders eliminates virtually all modelled preventable tail loss under base assumptions. Targeted interventions (fast lanes, dedicated safety stock for top 1% value SKUs) deliver modelled ROI in simulation. These are **hypotheses for further investigation** with real WMS data, not operational prescriptions.

7. Limitations & Assumptions (Full Disclosure)

This section is intentionally prominent. Every limitation below was known and accepted at the outset due to undergraduate resource constraints. The project prioritises transparency over over-claiming.

- **Data vintage:** 2010–2011 transactional data. E-commerce, fulfilment technology, customer expectations, carrier networks, and WMS capabilities have transformed since then. Results may not generalise to 2026 operations.
- **Fully synthetic delays:** No real warehouse telemetry, no actual SLA breach timestamps, no real dragon-order fulfilment times. All delay distributions, surge probabilities, and dragon bias are simulated from assumed parameters.
- **Parameter sourcing:** All numeric inputs (dragon frequency 0.025–0.032%, delay means 360–420 min, holding rate 25%, bias exponents, sigmas, clip ranges, SLA thresholds) are educated guesses derived from publicly available summaries and secondary sources, not the full 2025 WERC/CSCMP proprietary datasets.
- **Monte Carlo assumptions:** Daily dragon revenue assumed i.i.d. lognormal for computational simplicity. Hawkes clustering is modelled separately and not yet integrated into the MC paths.
- **Causal inference shortcuts:** Propensity-score matching uses standard 1:1 caliper without balance diagnostics or double-robustness. Quantile regression run on single subsample (now with 500 bootstrap replications).
- **Backtesting power:** Only ~374 days of data split into purged expanding windows (180-day min train, 30-day step, 14-day purge). EVT tail modelling and Hawkes fitting on full sample (overfitting risk). Statistical power to detect model mis-calibration is low.
- **BSTS priors:** Bayesian Structural Time Series uses conservative fixed Q/R priors rather than full hierarchical Bayesian updating.
- **GPD/Hawkes fitting:** Extreme-value and self-exciting models fitted on full sample rather than rolling or purged windows.
- **No production validation:** No A/B test, no real-time WMS integration, no out-of-sample live data. All results are in-sample simulation outputs.
- **Financial extrapolation:** 5-year cumulative figures assume constant dragon frequency, holding rates, and SLA policies — unrealistic in dynamic retail environments.

8. What This Project Demonstrates vs. What It Does Not Claim

Demonstrates (for a 20-year-old undergraduate with public-data constraints):
- End-to-end implementation of a non-trivial quantitative pipeline (EVT/GPD, Monte Carlo, Hawkes, BSTS, PSM + quantile regression, purged backtesting).
- Clean, modular Python architecture (src/ with subpackages: data/, delay_simulation/, causal_engine/, risk/, backtest/, hwk_bsts_forecasting/, utils/, config.py).
- Rigorous documentation of every assumption and limitation.
- Professional reproducibility (requirements.txt, pyproject.toml, run instructions, Kaggle notebooks).

Does NOT claim or demonstrate:
- A validated operational model ready for warehouse use.
- Real-world financial impact figures (all £ numbers are simulation outputs under assumed parameters).
- Evidence that 99th-percentile SLA on dragons is the correct policy (this is a hypothesis generated by the model, not a tested conclusion).
- Generalisability beyond the 2010–2011 data and synthetic delay assumptions.

9. Repository Contents

- **src/** — Complete Python pipeline (data ingestion → preprocessing → EVT → delay simulation → MC VaR/ES → causal validation → Hawkes + BSTS → purged backtesting → reporting).
- **notebooks/** — Executable Python scripts for Scenario 1, Scenario 2, and full pipeline runner (run_src.py).
- **dataset/** — Raw and cleaned UCI Online Retail data (public).
- **results/** — VaR/ES outputs, backtesting diagnostics, sensitivity tables, plots.
- **reports/** — Full technical documentation and dashboards.
- **config.py** — All parameters, backtest settings, causal hyperparameters in one place.
- **LIMITATIONS.md** — Full disclosure of assumptions and constraints.
- **README.md + README_reproducibility_section.md** — Execution instructions and methodology overview.

10. Technologies

Core: Python 3, pandas, NumPy, SciPy, statsmodels
Advanced: PyMC / PyStan (BSTS), custom Hawkes process implementation, scikit-learn (PSM + quantile regression), extreme-value modelling libraries
Visualisation: Plotly, Matplotlib, Seaborn
Configuration: pyproject.toml, requirements.txt

11. References

1. Chen, D. (2015). Online Retail [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5BW33 — *Primary transactional dataset (2010–2011). Used as-is; no newer public equivalent available.*

2. Warehousing Education and Research Council (WERC). (2025). DC Measures Report 2025. — *Parameters informed by publicly available summaries and secondary sources (Yale Materials Handling infographic, press releases). Full proprietary report unavailable.*

3. Council of Supply Chain Management Professionals (CSCMP) & Kearney. (2025). State of Logistics Report 2025. — *Macro-level volatility and resilience metrics from public summaries only.*

4. Yale Materials Handling. (2025). Top 12 distribution center metrics [Infographic based on WERC 2025]. https://www.yale.com/... — *Public secondary source used for best-in-class benchmarks (on-time ≥99.5%, dock-to-stock <3.5h).*

FINAL DISCLAIMER

This document and the associated repository represent an undergraduate research exercise in quantitative methods applied to supply-chain tail risk. The author makes no claim that the numerical results are accurate, generalisable, or suitable for decision-making in any real warehouse or retail operation. Any resemblance to operational advice is coincidental and unintended. Users should treat all financial estimates as illustrative Monte Carlo outputs under arbitrary assumptions. Real validation requires proprietary WMS data, live A/B testing, and domain expertise the author does not possess.